Project 3
Small Perturbation Angle Stability

Figure 1 shows the WSCC 9-bus test system employed for the proposed project. The impedances of the lines and the reactances of the generators are given in a common per-unit base. The loads are modelled as constant impedances and given in per unit. The frequency of the grid is 50 Hz. The grid is implemented in PowerWorld and MatLab; all the files can be found in the folder provided on WeBeep.

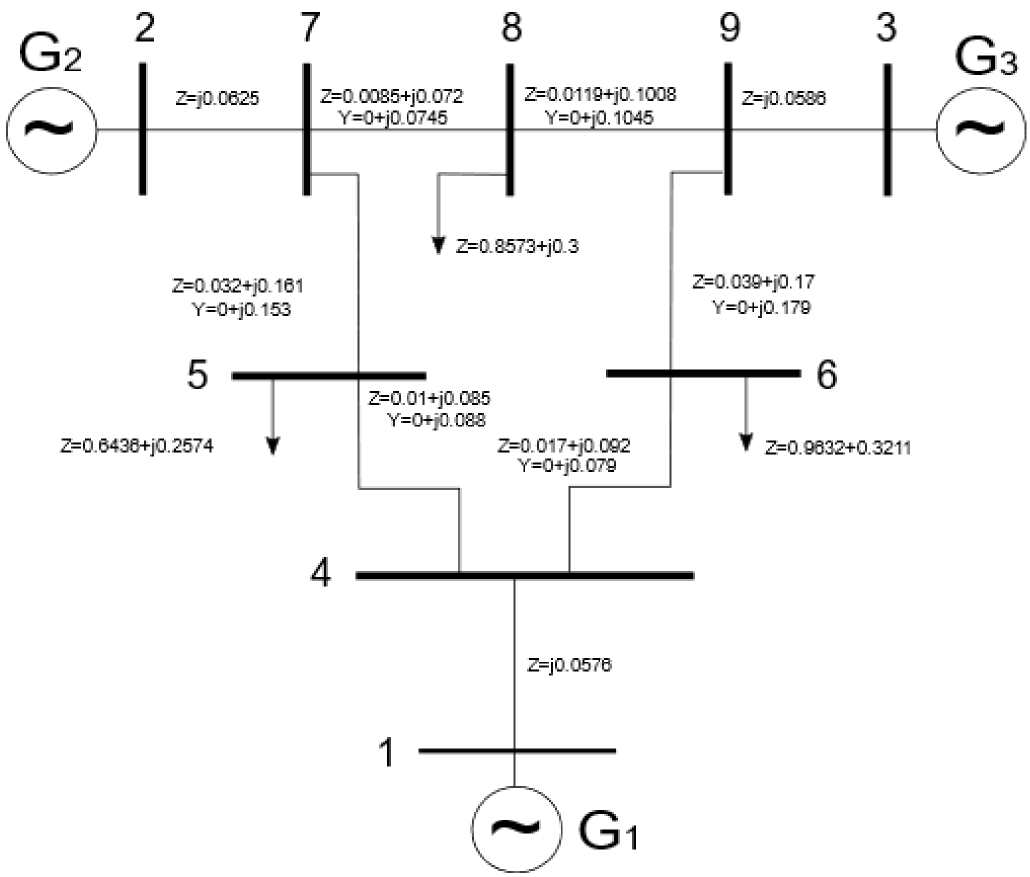

Figure 1. WSCC test system.

Consider the fourth-order model for the generators, where the electrical torque (T_{ele}) can be expressed as a function of fluxes (ψ) and currents (I) by:

$$\psi_q = \psi_q'' - I_q X_d''$$

$$\psi_d = \psi_d'' - I_d X_d''$$

$$T_{ele} = \psi_d I_q - \psi_q I_d$$

The swing equation, according to the PowerWorld notation, is (ω_0 in [rad/s]):

$$\dot{\delta} = \Delta\omega * \omega_0$$

$$\dot{\omega} = \frac{1}{2H}\left(\frac{P_{mec} - D * \Delta\omega}{1 + \Delta\omega} - T_{ele}\right)$$

The machine parameters are listed in Table 1:

Table 1. Machine data.

Parameter	G1	G2	G3
H [s]	25.32	5.2	2.98
Xd [p.u.]	0.146	0.8958	1.3125
X'd [p.u.]	0.0608	0.1198	0.1813
X''d [p.u.]	0.0400	0.0900	0.1500
Xq [p.u.]	0.0969	0.8645	1.2578
X'q [p.u.]	0.0969	0.1969	0.25
Xl [p.u.]	0.0300	0.0600	0.1200
T'do [s]	8.960	6.000	5.890
T''do [s]	0.030	0.030	0.030
T'qo [s]	0.310	0.535	0.600
T''qo [s]	0.050	0.050	0.050

All the generators are equipped with the same exciter (IEEET1 – type 1) and Power System Stabilizer (PSS1A – The input signal is the rotor speed deviations in p.u.), whose parameters are shown in Table 2 and Table 3; the corresponding block diagrams are reported in Figure 2 and Figure 3:

Table 2. Exciter parameters.

KA	20
TA [s]	0.2
KE [p.u.]	1.0
TE [s]	0.314
KF [p.u.]	0.063
TF [s]	0.35
E1	2.8
E2	3.73
SE1	0.30338
SE2	1.28840

with: $S_{Ei}(E_{fdi}) = A * e^{B*E_{fdi}}$, where: $A = \frac{SE_1}{e^{B*E_1}} = 0.0039$ and $B = \frac{\ln\left(\frac{SE_1}{SE_2}\right)}{E_1 - E_2} = 1.555$.

Table 3. PSS parameters.

T1 [s]	0.5
T2 [s]	0.01
T3 [s]	0.5
T4 [s]	0.01
T5 [s]	10
Ks	0.1
T6 [s]	0.02
A_1=A_2	0

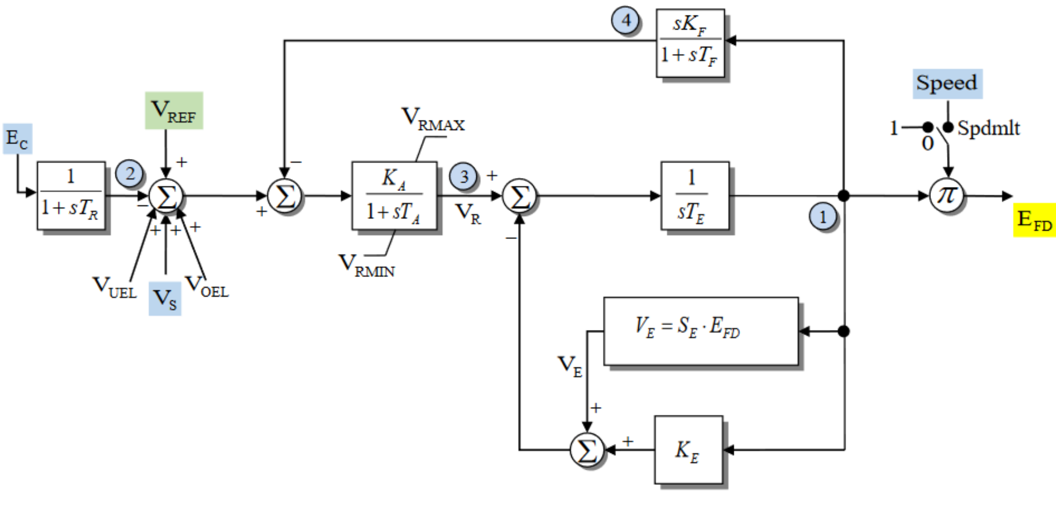

Figure 2. Exciter block diagram.

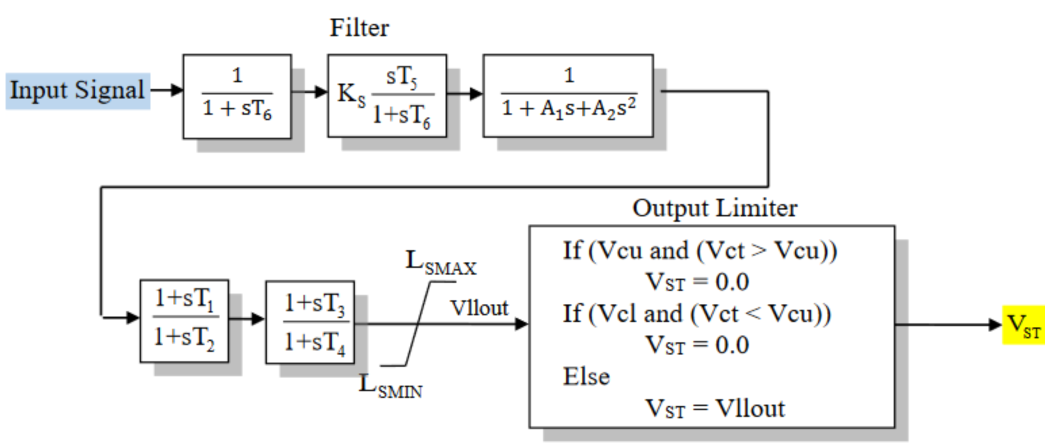

Figure 3. PSS block diagram.

Using the Matlab script for the modal analysis and PowerWorld for the transient analysis, answer the following points:

- With all the exciters and all the PSS switched off:
    a. Plot the eigenvalues in the complex space.
       How many electromechanical modes have you obtained? Provide the damping ratio and the frequency of the electromechanical modes identified.
       Which generator oscillates against which? Can you classify the modes? Analyze the results by plotting the mode shapes.
       Can you classify the remaining non-electromechanical, oscillatory modes?
    b. After a transient (line opening), the speeds of the generators are altered. Plot the time domain behavior in PowerWorld: is it coherent with your modal analysis?
- Switch on all the exciters and redo the analysis of point 1:
    a. Do you get any improvement in the damping ratios of the electromechanical modes? Comment the results.
- Switch on also all the PSS and redo the analysis of point 1:
    a. Do you get an improvement in the damping ratios of the electromechanical modes?
    b. Change the gain of the PSS (K_S) and try to improve the damping of the electromechanical mode. Comment the results.